

DSA STUDY BLUEPRINT SERIES

09. Pointers in CPP In Detail

Memory Addresses, Dereferencing, and Pointer Arithmetic

Section 02 • C++ Core Data Structures & Algorithms

Core Concepts & Visual Blueprint

Theoretical models and memory architectures

Key Principles

- **Pointers:** Variables that store the memory address of another variable.
- **Address-of (&):** Retrieves the memory address.
- **Dereference (*):** Accesses the value stored at that address.

Memory & Execution Blueprint



C++ Implementation Reference

Standard code layout and syntax

```
int a = 10;
int* ptr = &a;
cout << "Address: " << ptr << endl;
cout << "Value: " << *ptr << endl;
*ptr = 20; // updates 'a' directly
```

Algorithmic Complexity & Best Practices

Performance evaluations and critical guidelines

Performance Table

Aspect / Case	Complexity
Pointer Dereference	$O(1)$
Space Complexity	$O(1)$

Key Practices & Gotchas

- **Important:** Always initialize unused pointers to `nullptr` to avoid garbage reference crashes.
- Verify boundary conditions (e.g. empty inputs, single elements).
- Optimize dynamic memory allocations to prevent leaks.

Dry Run & Step-by-Step Execution

Trace analysis on a sample input case

Execution Walkthrough

Let's dry run the address assignments and pointer dereferences in memory:

Step / Instruction	Variable 'a' Value	Variable 'a' Address	Pointer 'ptr' Value	Deref '*ptr' Value
int a = 10;	10	0x7ffe100	Uninitialized	N/A
int* ptr = &a;	10	0x7ffe100	0x7ffe100	10
*ptr = 20;	20 (Updated!)	0x7ffe100	0x7ffe100	20

Interview Variations & Optimization Pathways

Common follow-up problems and optimization strategies

Key Variations & Follow-up Questions

- **Pointer to Pointer:** Multiple indirection levels (e.g. `int** ptr`) store memory locations of pointers, forming hierarchical memory reference grids.
- **References vs Pointers:** References are compiler aliases that must be bound at definition and cannot be re-assigned or set to null, whereas pointers are re-assignable and support arithmetic.
- **Gotcha:** Freeing memory via `delete` does not change the pointer value. Set freed pointers to `nullptr` immediately to avoid dangling pointer references.
- **Related LeetCode Problems:** #141 Linked List Cycle, #19 Remove Nth Node From End of List.